

Training Load, Neuromuscular Readiness, and Perceptual Fatigue Profile in Youth Elite Long-Jump Athletes

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Purpose: To describe and identify individual trends and changes in training load, neuromuscular readiness, and perceptual fatigue measures in 2 youth elite long jumpers, finalists at the European Athletics U18 (Under 18) Championships (athlete A, female, age 16.5 y, long-jump record 6.25 m; athlete B, male, age 16.0 y, long-jump record 7.28 m). **Methods:** Data were collected from both training sessions and athletics competitions during a 16-week period, divided into a preparation (weeks 1–8) and a competitive phase (weeks 9–16). Training load was computed through training diaries (training time, sprint, jumping, and weights volume). The countermovement jump and the 10-to-5 repeated-jump test were executed on a weekly basis to assess neuromuscular readiness, and perceptual fatigue measures were collected through a wellness questionnaire. Statistical analysis was conducted using a magnitude-based decisions approach. **Results:** The results highlighted a decrease in training load during the competitive period with *moderate to large* differences for training time, sprint, and jump volume. Moreover, data showed an upward trend and *very likely* higher scores in vertical-jump performance across the competitive phase together with a *very likely* lower perceptual fatigue. **Conclusions:** This scenario seemed to be favorable to achieve competition performance very close to the personal record during the competitive season. This study provided an example of application of a comprehensive monitoring system with young athletes involved in track-and-field jumping events.

Keywords: athletics, monitoring, elite athlete

Tracking performance development in young athletes represents a paramount aspect in creating a safe and successful environment for the growth and development of sport talent. In most track-and-field events, peak performance of world-class athletes is achieved at 25–27 years of age.¹ During adolescence, athletes' competition results are substantially improved,^{2,3} meanwhile, over the same period, they are exposed at a greater injury risk if they are training with excessively high volumes.⁴ Considering the described scenario, training and competition data can be used to quantify the load imposed, the athlete's readiness, and the risk associated with changes in load, with the ultimate goal of assisting the coaches in their decision-making process.⁵ Despite the growing interest on athlete monitoring systems, there is a paucity of real examples of the preparation of elite athletes in individual sports.⁶ The objectives of this study were to describe and identify individual trends and changes in training load, neuromuscular readiness, and perceptual fatigue measures in 2 youth elite long jumpers, finalists at the European Athletics U18 Championships.

Methods

Subjects

One female young jumper (athlete A: age 16.5 y, height 1.67 m, weight 59.0 kg, long jump personal record 6.25 m) and one male young jumper (athlete B: age 16.0 y, height 1.83 m, weight

69.0 kg, long jump personal record 7.28 m), representative of an athletics academy, agreed to participate in the study. Participants competed at the national and international level during the 2018 season and qualified for the European Athletics U18 Championships, where both achieved the final in the women's (athlete A: eighth place) and men's (athlete B: second place) long jump events, respectively. Written informed consent was obtained from participants, along with parental consent. The study is conformed to the recommendation of the Declaration of Helsinki.

Design

The study described herein was a case study.

Methodology

The 2 athletes were monitored during an athletics outdoor season for a period of 16 weeks, divided into 2 phases: a preparatory (from week 1 to week 8) and a competitive period (from week 9 to week 16).

Training load was quantified as external workload through detailed training diaries indicating weekly training volumes (training time, sprint, jumping, and weights). Sprint training (in meters) implies all forms of acceleration and sprint run with maximal effort, while jumping training (n) refers to all forms of jumping movements, such as in-place jumps, short bounds, extended bounds, depth jumps, and discipline-specific jumps. At the end, weights training (in kilograms) estimates the quantification of workload lifted during the strength sessions using the formula: number of sets $\times$ number of repetitions $\times$ weight lifted.

On a weekly basis, the countermovement jump and the 10-to-5 repeated jump test were executed using the OptoJump Next (Microgate, Bolzano, Italy) to assess neuromuscular readiness.

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Weekly testing measurements were performed on the first training session of the week ensuring 48 hours of recovery from the last training session and/or competition. Countermovement jump was performed executing 2 trials separated by 2-minute passive rest according to a previously outlined procedure. Jump height was reported in centimeters and used for the subsequent analyses as average value. The 10-to-5 repeated jump test involved participants performing optimal vertical rebounds (ie, maximal elevation at each jump) with minimal ground contact (ie, <0.250 ms) performed for a set of 11 jumps. The instructions given to the athletes were to “maximize jump height” and “minimize ground contact time.” The 5 jumps with the greatest reactive strength index (ie, the ratio of jump height to contact time) were averaged and used for further analyses. These assessments have been previously validated for the purpose of monitoring neuromuscular readiness in the athletes.^{7,8}

Perceptual fatigue measures were collected daily through a wellness questionnaire filled out by the athletes each morning. Using a 7-point Likert scale (eg, 0 = no soreness to 6 = extremely sore), the questionnaire describes the perceptual level of fatigue. The questionnaire gathers ratings of perceived fatigue, delayed-onset muscle soreness, health, sleep quantity, sleep quality, and mood, and then calculates the total score.^{9,10} Weekly average values for the total perceptual fatigue were used for the analyses.

Finally, results obtained in official competitions during the competitive phase were reported as relative performance to personal record (in percentage).

Statistical Analysis

Data are expressed as mean (SD). Differences between preparatory and competitive period in training load variables were computed using a magnitude-based decisions approach.¹¹ Quantitative chances of real differences in variables were assessed qualitatively as <1%, almost certainly not; <5%, very unlikely; <25% unlikely, probably not; 25% to 75%, possibly, possibly not; >75%, likely, probably; >95%, very likely; and >99.5%, almost certainly. If the chances of a variable having higher and lower differences were >5%, the true effect was deemed to be unclear. Otherwise, we interpreted that change as the observed chance. Thresholds for effect size statistics were interpreted as <0.2, trivial; 0.2 to 0.6, small; 0.6 to 1.2, moderate; 1.2 to 2.0, large; and >2.0, very large.¹² Changes in neuromuscular readiness test and perceptual fatigue measures were also analyzed using magnitude-based decisions. Linear trend and changes were assessed against the coefficient of variation and the typical error on a customized spreadsheet.¹³ The smallest worthwhile change was calculated as 0.3 times the within-subject SD. Smallest worthwhile change, typical error, and coefficient of variation were calculated for each athlete. To assess the changes from linear trend, data from the 4 weeks preceding the data collection were averaged and included to fit a straight-line trend (baseline) to calculate changes during the season.

Results

Table 1 reports the differences in external training load data between the preparatory and the competitive period for the athletes A and B. Individual trend and changes in countermovement jump, 10-to-5 repeated jump test, and perceptual total fatigue are depicted

in Figure 1, together with the relative competition result to personal record (in percentage).

Discussion

The main findings of this case study highlighted a substantial decrease in training load during the competitive period and an increasing trend in vertical jump performances across the competitive phase, together with a diminished perceptual fatigue.

As the competition phase approached, the training load decreased in terms of training time, sprint, and jump volume. A reduction in workload during the most important phases (ie, competitive and peaking) of the season is required to achieve peak performance in individual sports and allows for the implementation of sport-specific training needed for competition preparation.^{14,15} This load pattern may have influenced the neuromuscular readiness status of the athletes. In this light, superior performances in the vertical jump height and reactive strength index were observed across the end of the competitive phase, with some peaks in the last weeks prior of a major championship. Accordingly, previous research investigated the correlation between vertical jump score and performance showing an increasing trend in the weeks preceding the season-best results in senior sprinting and jumping athletes.^{7,16}

Moreover, the weekly perceptual total fatigue showed an opposite development displaying substantial reduction and improvements in subjective well-being over the competitive period. This outcome is in line with the other monitoring metrics used in the study and confirms the importance of frequently monitoring the stress–recovery balance in young athletes due to their extratraining activities (eg, school, lifestyle, relationship) with the goal to manage athlete preparation.¹⁷

Practical Applications

The findings from this case study suggest how training prescription, in terms of external loads, can be made in accordance with the period of the season and the distance from major competitions. In addition to workload monitoring, weekly vertical jump tests and daily wellness questionnaire can serve as practical tools to better understanding an athlete’s readiness to train and compete. Track-and-field coaches can use training load, neuromuscular readiness, and perceptual fatigue measures to monitor an athlete’s development and consequently adjust training contents using more information and, ultimately, maximizing performance results.

Conclusions

A reduction in training time and external training volume (sprint and jumps), together with an upward trend in vertical jump performance and a diminished perceptual fatigue during the competitive period, seemed to be favorable in maximizing performance in elite young jumpers competing at an international level. This study provided an example of an application of a comprehensive monitoring system with 2 young athletes involved in track-and-field jumping events, providing objective measures of external load during an extensive period of time.

Table 1 Differences in External Training Load Between the Preparatory and Competitive Periods

Athlete	External training-load variable	Mean (SD)		Mean difference (90% CI)	Standardized difference (90% CI)	Magnitude-based decision
		Preparatory	Competitive			
A	Training time, h	6.75 (2.38)	4.88 (2.59)	-1.9 (-4.1 to 0.3)	-0.70 (-1.52 to 0.12) moderate	Likely ↓ (4/11/85)
	Sprint volume, m	453.75 (253.66)	263.75 (157.93)	-190.0 (-379.7 to -0.3)	-0.67 (-1.33 to 0.00) moderate	Likely ↓ (2/10/88)
	Jumps volume, n	102.5 (82.81)	36.88 (27.89)	-65.6 (123.1 to -8.2)	-0.70 (-1.32 to -0.09) moderate	Likely ↓ (1/7/92)
	Weights volume, kg	893.75 (1306.42)	995.00 (897.33)	101.3 (-897.5 to 1100.0)	0.07 (-0.61 to 0.75) trivial	Unclear (37/38/25)
B	Training time, h	7.50 (1.41)	5.25 (2.31)	-2.3 (-4.0 to -0.5)	-1.41 (-2.50 to -0.33) large	Very likely ↓ (1/2/97)
	Sprint volume, m	550.00 (243.49)	235.00 (151.75)	-315.0 (-497.2 to -132.8)	-1.15 (-1.81 to -0.48) moderate	Very likely ↓ (0/1/99)
	Jumps volume, n	121.88 (81.33)	56.25 (28.63)	-65.6 (-122.3 to -8.9)	-0.72 (-1.34 to -0.10) moderate	Likely ↓ (1/7/92)
	Weights volume, kg	1505.00 (1904.00)	1722.50 (1164.37)	217.5 (-1200.1 to 1635.1)	0.10 (-0.56 to 0.76) trivial	Unclear (40/39/22)

Abbreviations: ↑, increase; ↓, decrease; CI, confidence interval.

Athlete A

Athlete B

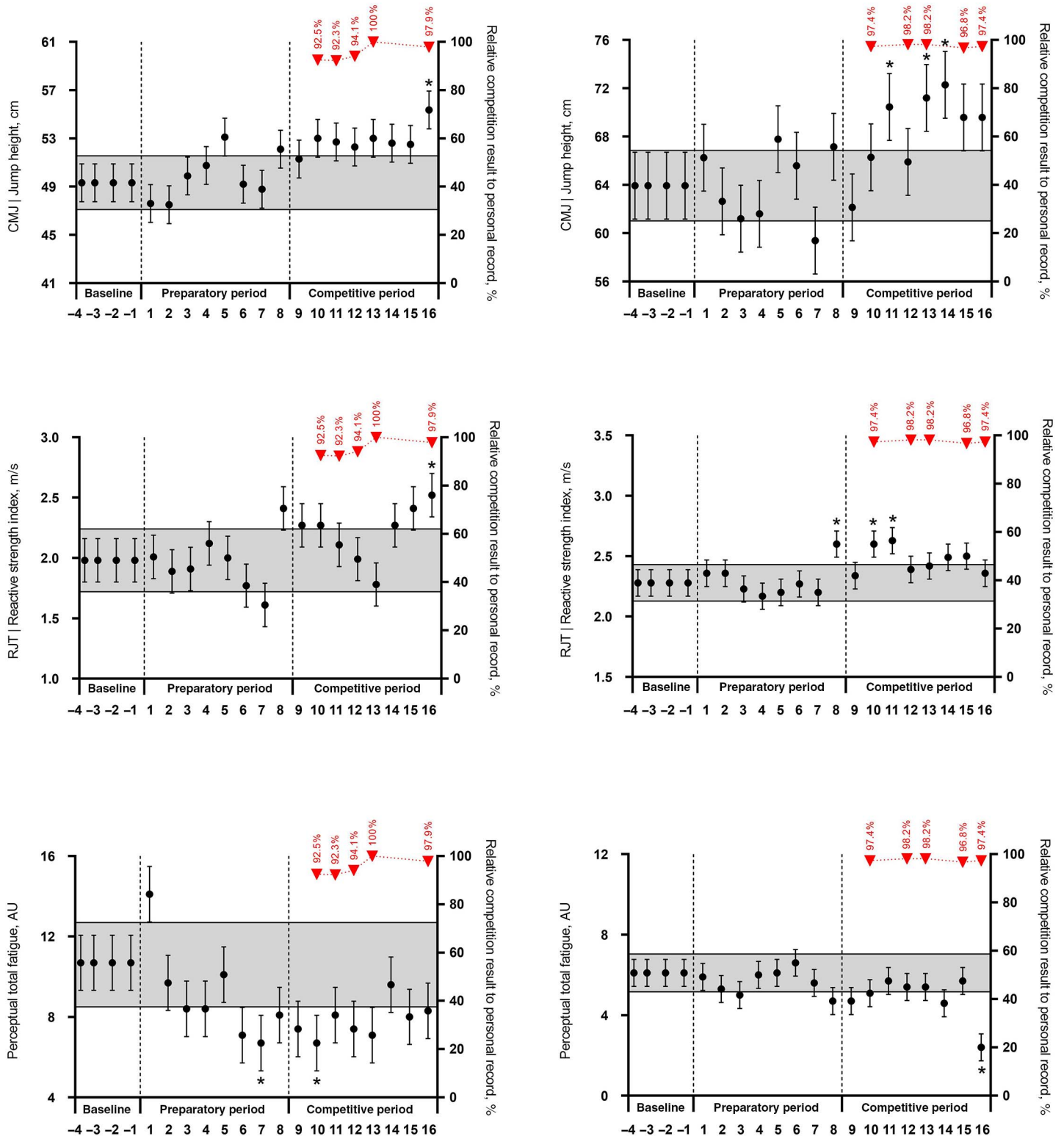


Figure 1 — Individual trend in CMJ, 10-to-5 RJT, and perceptual total fatigue for athletes A and B. Gray area represents the coefficient of variation to assess meaningful changes during the outdoor season from baseline. Black dots depict the weekly selected measures ($\pm$ typical error of measurement) during the preparatory and competitive periods (*very likely). Upside-down triangles depict the weekly competition result expressed as percentage of the personal record. AU indicates arbitrary units; CMJ, countermovement jump; RJT, repeated-jump test.

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